

Paper Club · Sep 9

Scientific Reviewer

8 + 3 minutes

Strengths · weaknesses · actionable ablations

arXiv 2512.06951 · 1st Place BEHAVIOR 2025

Seat 4

Strengths

Code + HF weights public

**Coherent correlated-noise
train ↔ infer story**

Honest: “not rigorous science”

pub $\approx$ priv (0.2605 / 0.2599)

System 2 for visual aliasing

**Competition-winning stack
with clear control spine**

Weaknesses

No factorial ablations

1st place $\neq$ component necessity

Task-ID + 4 ckpts

weakens open-vocab VLA claim

Heuristics mixed

science vs engineering wrappers

Underspecified

LR / seeds / curves missing

Gripper 2.2×

only $13 \times 3 = 39$ episodes

Undertrained (~2 ep)

confounds architecture claims

Actionable Ablation List

P0

Gripper OFF · System2 OFF · E1×E2 factorial

P1

Soft inpaint modes · compression · exec horizon 10-26 · 1 vs 4 ckpts

P2

$\beta \in \{0, 0.5, 1\}$ retrain · attn freeze · multi-sample N · FAST λ

Report

Seeds + learning curves · separate heuristics from methods tables

Prioritized Experiments

P0

E1

Gripper-rule OFF

P0

E2

System 2 voting OFF

P0

E3

E1×E2 factorial ← talk figure

P1

E4-E7

Inpaint / compression / horizon / 1 vs 4 ckpts

P2

E8-E12

Retrain β / attn / multi-sample / FAST / full 50

Reviewer close

Strengths real · science incomplete without factorials